

Math 514, F13
Quiz 1

Name: Answer
SSN: xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) If two balls are withdrawn from a bag that contains 9 red balls and 6 blue ones.

(a) What is the probability that both balls are red?

$$P\{(R,R)\} = \frac{9}{15} \cdot \frac{8}{14} = \frac{12}{35}$$

(b) What is the probability that one is red and the other is blue?

$$P\{(R,B), (B,R)\} = \frac{9}{15} \cdot \frac{6}{14} + \frac{6}{15} \cdot \frac{9}{14} = \frac{18}{35}$$

Problem 2. (5 points) Four buses carry 28, 19, 22 and 31 passengers respectively. One of the total 100 passengers is randomly chosen and let X denote the number of passengers on the bus of the selected passengers.

(a) Find $P\{X > 25\}$.

$$S = \{\text{Bus 1}, \text{Bus 2}, \text{Bus 3}, \text{Bus 4}\}$$

$$P\{X > 25\} = P\{\text{Bus 1}, \text{Bus 4}\} = \frac{28}{100} + \frac{31}{100} = \frac{59}{100}$$

(b) Find $E[X]$.

$$\begin{aligned} E[X] &= \sum_{i=1}^4 x_i \cdot P\{X = x_i\} \\ &= 28 \cdot \left(\frac{28}{100}\right) + 19 \cdot \left(\frac{19}{100}\right) + 22 \cdot \left(\frac{22}{100}\right) + 31 \cdot \left(\frac{31}{100}\right) \\ &= \frac{2590}{100} \\ &= 25.9 \end{aligned}$$